Small Signal Schottky Diode



FEATURES

- For general purpose applications
- This diode features very low turn-on voltage and fast switching
- This device is protected by a PN junction guard ring against excessive voltage, such as electrostatic discharges
- AEC-Q101 qualified available
- Base P/N-E3 - RoHS-compliant, commercial grade
- Base P/N-HE3 - RoHS-compliant, AEC-Q101 qualified
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

Pb-free

e3

RoHS COMPLIANT

DESIGN SUPPORT TOOLS

click logo to get started

3D Models Available

MECHANICAL DATA

Case: SOD-123

Weight: approx. 10.3 mg

Packaging codes/options:

18/10K per 13" reel (8 mm tape), 10K/box

08/3K per 7" reel (8 mm tape), 15K/box

PARTS TABLE				
PART	ORDERING CODE	CIRCUIT CONFIGURATION	TYPE MARKING	REMARKS
BAT46W	BAT46W-E3-08 or BAT46W-E3-18	Single	L6	Tape and reel
	BAT46W-HE3-08 or BAT46W-HE3-18			

ABSOLUTE MAXIMUM RATINGS (T_{amb} = 25 °C, unless otherwise specified)				
PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Repetitive peak reverse voltage		V_{RRM}	100	V
Forward continuous current (1)		I_F	150	mA
Repetitive peak forward current (1)	t_p < 1 s, δ < 0.5	I_{FRM}	350	mA
Surge forward current (1)	t_p < 10 ms	I_{FSM}	750	mA
Power dissipation (1)	T_{amb} = 65 °C	P_{tot}	150	mW

Note

(1) Valid provided that electrodes are kept at ambient temperature

THERMAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)				
PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Thermal resistance junction to ambient air (1)		R_{thJA}	300	K/W
Junction temperature		T_j	125	°C
Operating temperature range		T_{op}	-55 to +125	°C
Storage temperature range		T_{stg}	-55 to +150	°C

Note

(1) Valid provided that electrodes are kept at ambient temperature

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ELECTRICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)						
PARAMETER	**TEST CONDITION**	**SYMBOL**	**MIN.**	**TYP.**	**MAX.**	**UNIT**
Reverse breakdown voltage	I_R = 100 µA (pulsed)	$V_{(BR)}$	100			V
Leakage current [1]	V_R = 1.5 V	I_R			0.5	µA
	V_R = 1.5 V, T_j = 60 °C	I_R			5	µA
	V_R = 10 V	I_R			0.8	µA
	V_R = 10 V, T_j = 60 °C	I_R			7.5	µA
	V_R = 50 V	I_R			2	µA
	V_R = 50 V, T_j = 60 °C	I_R			15	µA
	V_R = 75 V	I_R			5	µA
	V_R = 75 V, T_j = 60 °C	I_R			20	µA
Forward voltage [1]	I_F = 0.1 mA	V_F			250	mV
	I_F = 10 mA	V_F			450	mV
	I_F = 250 mA	V_F			1000	mV
Diode capacitance	V_R = 0 V, f = 1 MHz	C_D		10		pF
	V_R = 1 V, f = 1 MHz	C_D		6		pF

Note
[1] Pulse test; $t_p \leq 300$ µs, $\delta < 2$ %

TYPICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)

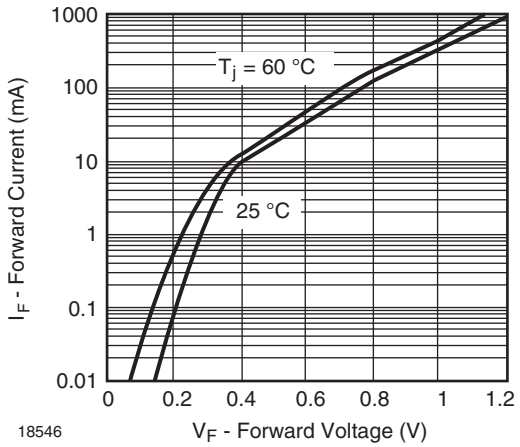


Fig. 1 - Typical Instantaneous Forward Characteristics

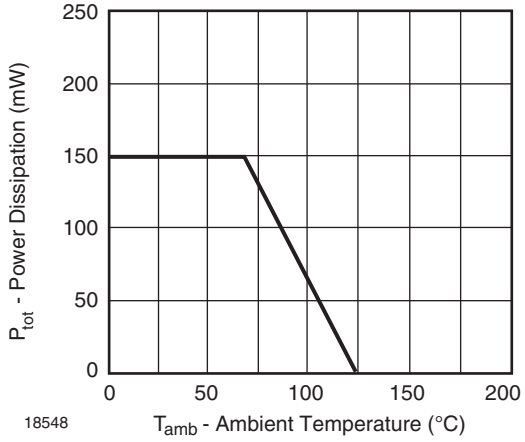


Fig. 3 - Admissible Power Dissipation vs. Ambient Temperature

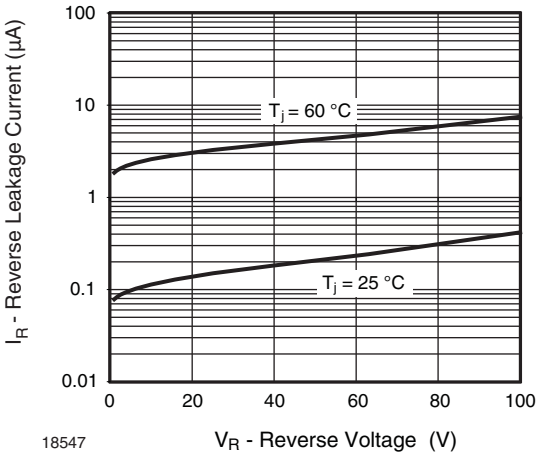
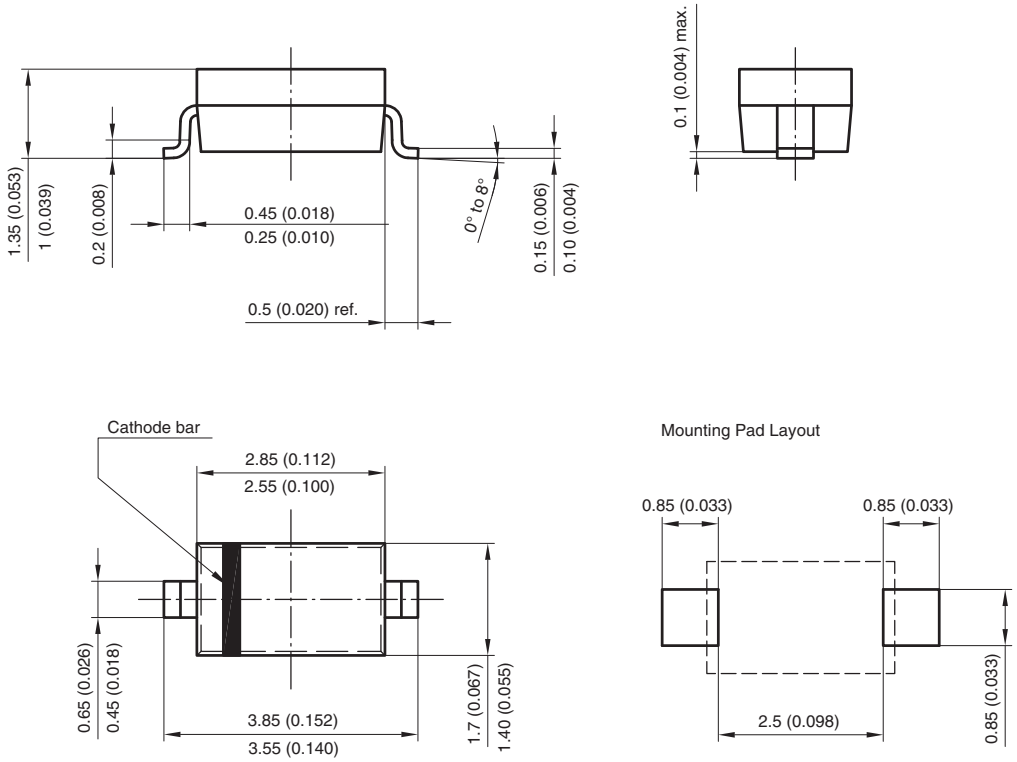


Fig. 2 - Typical Reverse Characteristics

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PACKAGE DIMENSIONS in millimeters (inches): **SOD-123**



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